

Numerical Analysis homework 1

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I. The width of bisection interval in different loops

I-a

It's easy to find that the width is reduced by half at each step, so:

$$width_n = (3.5 - 1.5)/2^n = \frac{1}{2^{n-1}}$$

I-b

If we set the δ as postcondition or in n th step, we know:

$$\sup |r - mid| \leq \delta/2 \quad \text{or} \quad \leq \frac{1}{2^n}$$

II. Prove the relative error of bisection no greater than ϵ in such condition

By I.b, we have: relative error $e = \frac{|mid-r|}{r} \leq \frac{b_0-a_0}{2^{n+1}(a_0)}$, in which r is root ($0 < a_0 \leq r$), easily to prove both sides are positive. So:

$$e \leq \epsilon \iff \frac{b_0 - a_0}{2^{n+1}a_0} \leq \epsilon \iff \log(b_0 - a_0) - (n+1)\log 2 - \log a_0 \leq \log \epsilon$$

i.e.

$$\log(b_0 - a_0) - \log a_0 - \log \epsilon \leq (n+1)\log 2$$

$$\frac{\log(b_0 - a_0) - \log a_0 - \log \epsilon}{\log 2} - 1 \leq n$$

□

III. Perform four iterations of Newton's method

Easily to acquire $p'(x) = 12x^2 - 4x$, by Newton's method formula:

$$x_{n+1} = x_n - \frac{p(x_n)}{p'(x_n)}$$

So

$$x_{n+1} = x_n - \frac{4x_n^3 - 2x_n^2 + 3}{12x_n^2 - 4x_n}$$

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Table 1: Result				
	x_1	x_2	x_3	x_4
Value(approximate)	-0.8125	-0.7708	-0.7689	-0.7688

IV. The error analysis of the variation of Newton's method

By Taylor's expansion around x_0 & α with Cauchy remainder

$$f(x) = f(\alpha) + f'(\alpha)(x - \alpha) + \frac{1}{2}f''(\eta)(x - \alpha)^2 = f'(\alpha)(x - \alpha) + \frac{1}{2}f''(\eta)(x - \alpha)^2$$

$$f(x) = f(x_0) + f'(x_0)(x - x_0) + \frac{1}{2}f''(\xi)(x - x_0)^2$$

so

$$f(x_n) = f'(\alpha)(x_n - \alpha) + \frac{1}{2}f''(\eta_n)(x_n - \alpha)^2 \Rightarrow \frac{f(x_n)}{f'(\alpha)} = x_n - \alpha + \frac{1}{2f'(\alpha)}f''(\eta_n)(x_n - \alpha)^2$$

$$0 = f(\alpha) = f(x_0) + f'(x_0)(\alpha - x_0) + \frac{1}{2}f''(\xi)(\alpha - x_0)^2$$

Therefore

$$x_{n+1} - \alpha = x_n - \frac{f(x_n)}{f'(x_0)} - \alpha = (1 - \frac{f'(\alpha)}{f'(x_0)})(x_n - \alpha) - \frac{1}{2f'(x_0)}f''(\eta_n)(x_n - \alpha)^2$$

i.e.

$$e_{n+1} = (1 - \frac{f'(\alpha)}{f'(x_0)})e_n - \frac{1}{2f'(x_0)}f''(\eta_n)e_n^2$$

Therefore

$$e_{n+1} = \begin{cases} Ce_n (s=1, C = (1 - \frac{f'(\alpha)}{f'(x_0)}) - \frac{1}{2f'(x_0)}f''(\eta_n)(x_n - \alpha)) & f'(\alpha) \neq f'(x_0), \\ Ce_n^2 (s=2, C = \frac{f''(\eta_n)}{2f'(x_0)}) & f'(\alpha) = f'(x_0). \end{cases}$$

V. The convergence of the iteration

Answer is Yes, we prove that $\{|x_n|\}$ deasing firstly, i.e.

$$|x_{n+1}| = |\arctan(x_n)| = \arctan(|x_n|) < |x_n|$$

By a trivail inequation $\tan x > x$ when $x \in [0, \frac{\pi}{2})$. Then we claim that $|x_n| \rightarrow 0$, if NOT:

$$\exists \epsilon_0 > 0 \quad \forall N > 0, \exists n > N \quad s.t. \quad \epsilon_0 \leq |x_n|$$

(actually $\forall n > N$ by deasing). Then let $f(x) = x - \arctan x$, so $f'(x) = 1 - \frac{1}{1+x^2} > 0 (\forall x > 0)$, So

$$\forall n > N, 0 < \epsilon_0 - \arctan(\epsilon_0) \leq |x_n| - |x_{n+1}|$$

so, after finite (at most $\lceil \frac{\epsilon_0}{\epsilon_0 - \arctan(\epsilon_0)} \rceil + 1 := s$) steps. We will get $|x_{n+s}| < \epsilon_0$. Contradiction! □

VI. The convergence of continued fraction

Attention that $x_{n+1} = \frac{1}{p+x_n}$, let $f(x) = \frac{1}{p+x}$, ($x > 0$). Then because $p > 1, x > 0 (x_n > 0)$:

$$\forall x, y > 0, |f(x) - f(y)| = |\frac{1}{(p+x)(p+y)}| |x - y| < \frac{1}{p^2} |x - y|$$

So $f(x)$ is a contraction. Therefore $\{x_n\}$ converge by the Theorem in course. We suppose the limit is x , then

$$\lim_{n \rightarrow \infty} x_{n+1} = \frac{1}{p + \lim_{n \rightarrow \infty} x_n} \Rightarrow x^2 + px - 1 = 0$$

Consider that x is positive, so $x = \frac{-p + \sqrt{p^2 + 4}}{2}$.

VII. The error of changed range in bisection algorithm

In this case, it's not appropriate to use relative error to measure. Because the root r may be zero, the fraction will be ∞ or undefined. Therefore we replace it with **absolute error**.

$$e \leq \epsilon \iff \frac{b_0 - a_0}{2^{n+1}} \leq \epsilon \iff \log(b_0 - a_0) - (n+1)\log 2 \leq \log \epsilon$$

i.e.

$$\begin{aligned} \log(b_0 - a_0) - \log \epsilon &\leq (n+1)\log 2 \\ \frac{\log(b_0 - a_0) - \log \epsilon}{\log 2} - 1 &\leq n \end{aligned}$$

VIII. Newton's method for solving multiplicity root

VIII-a

Method1:

In Newton's method

$$0 = f(\alpha) = f(x_n) + f'(x_n)(\alpha - x_n) + \frac{1}{2}f''(\xi_n)(\alpha - x_n)^2$$

So

$$\begin{aligned} x_{n+1} - \alpha &= x_n - \alpha - \frac{f(x_n)}{f'(x_n)} = \frac{1}{2f'(x_n)}f''(\xi_n)(x_n - \alpha)^2 \\ 2f'(x_n)e_{n+1} &= f''(\xi_n)e_n^2 \\ \frac{2f'(x_n)}{f''(\xi_n)}e_{n+1} &= \frac{f''(\xi_n)}{f''(x_n)}e_n^2 \end{aligned}$$

we let $n \rightarrow \infty$ and use L'Hopital's rule:

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{e_{n+1}}{e_n^2} &= \lim_{n \rightarrow \infty} \frac{f''(\xi_n)}{f''(x_n)} \cdot \frac{2f'(x_n)}{f''(x_n)} = 1 / \frac{f^{(k-1)}(\alpha)}{f^{(k)}(\alpha)} = \infty \\ \lim_{n \rightarrow \infty} \frac{e_{n+1}}{e_n} &= C \end{aligned}$$

Therefore, we can examine the convergence rate of sequence $\{x_n\}$.

Method2:

we define $Q_n := \frac{|x_{n+1} - x_n|}{|f(x_n)|}$, the similar way to use L'Hopital's rule. If its value be stable at some constant, it's the simple root. Otherwise, it be greater and greater, then it's multiplicity root.

VIII-b

Given function $g(x) = \frac{f(x)}{(x-\alpha)^k}$, we can give $g(x)$ a definition at α by analytical extension. So well-defined.

$$\begin{aligned} \frac{e_{n+1}}{e_n} &= 1 - k \frac{g(x_n)(x_n - \alpha)^{k-1}}{k(x_n - \alpha)^{k-1}g(x_n) + (x_n - \alpha)^k g'(x_n)} = \frac{(x_n - \alpha)^k g'(x_n)}{k(x_n - \alpha)^{k-1}g(x_n) + (x_n - \alpha)^k g'(x_n)} \\ &= \frac{(x_n - \alpha)g'(x_n)}{kg(x_n) + (x_n - \alpha)g'(x_n)} \end{aligned}$$

So

$$\lim_{n \rightarrow \infty} \frac{e_{n+1}}{e_n^2} = \lim_{n \rightarrow \infty} \frac{g'(x_n)}{kg(x_n) + (x_n - \alpha)g'(x_n)} \neq 0$$

□